

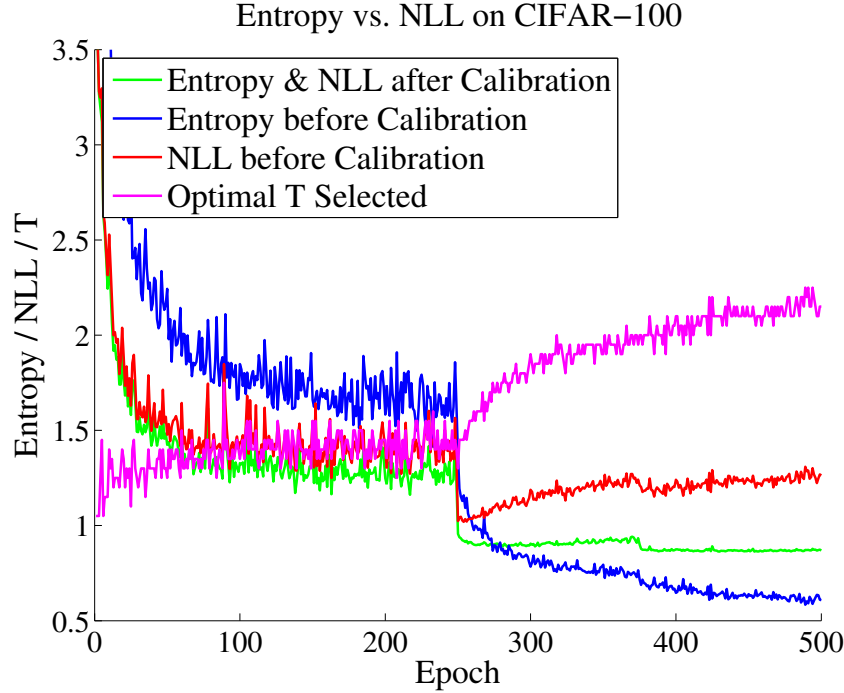


Figure S1. Entropy and NLL for CIFAR-100 before and after calibration. The optimal T selected by temperature scaling rises throughout optimization, as the pre-calibration entropy decreases steadily. The post-calibration entropy and NLL on the validation set coincide (which can be derived from the gradient optimality condition of T).

Dataset	Model	Uncalibrated	Hist. Binning	Isotonic	BBQ	Temp. Scaling	Vector Scaling	Matrix Scaling
Birds	ResNet 50	30.06%	25.35%	16.59%	11.72%	9.08%	9.81%	38.67%
Cars	ResNet 50	41.55%	5.16%	15.23%	9.31%	20.23%	8.59%	29.65%
CIFAR-10	ResNet 110	33.78%	26.87%	7.8%	72.64%	8.56%	27.39%	22.89%
CIFAR-10	ResNet 110 (SD)	34.52%	17.0%	16.45%	19.26%	15.45%	15.55%	10.74%
CIFAR-10	Wide ResNet 32	27.97%	12.19%	6.19%	9.22%	9.11%	4.43%	9.65%
CIFAR-10	DenseNet 40	22.44%	7.77%	19.54%	14.57%	4.58%	3.17%	4.36%
CIFAR-10	LeNet 5	8.02%	16.49%	18.34%	82.35%	5.14%	19.39%	16.89%
CIFAR-100	ResNet 110	35.5%	7.03%	10.36%	10.9%	4.74%	2.5%	45.62%
CIFAR-100	ResNet 110 (SD)	26.42%	9.12%	10.95%	9.12%	8.85%	8.85%	35.6%
CIFAR-100	Wide ResNet 32	33.11%	6.22%	14.87%	11.88%	5.33%	6.31%	44.73%
CIFAR-100	DenseNet 40	21.52%	9.36%	10.59%	8.67%	19.4%	8.82%	38.64%
CIFAR-100	LeNet 5	10.25%	18.61%	3.64%	9.96%	5.22%	8.65%	18.77%
ImageNet	DenseNet 161	14.07%	13.14%	11.57%	10.96%	12.29%	9.61%	-
ImageNet	ResNet 152	12.2%	14.57%	8.74%	8.85%	12.29%	9.61%	-
SVHN	ResNet 152 (SD)	19.36%	11.16%	18.67%	9.09%	18.05%	30.78%	18.76%
20 News	DAN 3	17.03%	10.47%	9.13%	6.28%	8.21%	8.24%	17.43%
Reuters	DAN 3	14.01%	16.78%	44.95%	36.18%	25.46%	18.88%	19.39%
SST Binary	TreeLSTM	21.66%	3.22%	13.91%	36.43%	6.03%	6.03%	6.03%
SST Fine Grained	TreeLSTM	27.85%	28.35%	19.0%	8.67%	44.75%	11.47%	11.78%

Table S1. MCE (%) (with $M = 15$ bins) on standard vision and NLP datasets before calibration and with various calibration methods. The number following a model's name denotes the network depth. MCE seems very sensitive to the binning scheme and is less suited for small test sets.

S4. Additional Reliability Diagrams

We include reliability diagrams for additional datasets: CIFAR-10 (Figure S2) and SST (Figure S3 and Figure S4). Note that, as mentioned in Section 2, the reliability dia-

grams do not represent the proportion of predictions that belong to a given bin.